Homework 2 due: Thursday 13$^{\text{th}}$ February, 2014

Choose any 10 problems from the following exercises in chapter 2 of the notes.

2.1 Suppose $f(x_1) = f(x_2)$ for any linear function $f\colon \mathbb{R} \to \mathbb{R}$ with a nonzero, defined slope. $f(x_1) = f(x_2) \Rightarrow mx_1 + b = mx_2 + b \Rightarrow x_1 = x_2$. So f is one-to-one.

2.2 Let $b \in \mathbb{Z}$. $f(b + 0.5) = \lfloor b + 0.5 \rfloor = b$. So f is onto. To see that f is not one-to-one, $f(1.5) = 1 = f(1)$ but $1.5 \neq 1$.

2.3 Recall that modulo operator distributes over addition and multiplication. So for any two integers a and b we have the following:

$f(a + b) = (a + b) \bmod n = (a \bmod n + b \bmod n) \bmod n = a \bmod n + b \bmod n = f(a) + f(a)$.

$f(ab) = (ab) \bmod n = ((a \bmod n)(b \bmod n)) \bmod n = a \bmod nb \bmod n = f(a)f(b)$.

So f is a homomorphism.

2.4 $f(1.5 + 1.5) = f(3) = 3 \neq 1 + 1 = f(1.5) + f(1.5)$

2.5 This problem should have read "determine if f is a homomorphism".

f is <u>not</u> a homomorphism under the usual operations since $f(2 + 3) = 2^{2+3} = 2^2 2^3 = 32 \neq 12 = 2^2 + 2^3 = f(2) + f(3)$. However it is a homomorphism under addition and multiplication defined in the range as follows: $2^x \oplus 2^y = 2^{x+y}$ and $2^x \otimes 2^y = 2^{xy}$. $f(x) \oplus f(y) = 2^{x+y} = f(x + y)$, and $f(x) \otimes f(y) = 2^{xy} = f(xy)$.

2.6 The key here is to look at how the identity is mapped.

The identity of $\mathbb{Z}_n$ is 1. Let $f\colon \mathbb{Z}_n \to \mathbb{Z}_n$ be a homomorphism, and suppose that $f(1) = a$ (1 must map to something). Since homomorphisms are operation preserving $f(x) = f(\underbrace{1 + 1 + \cdots + 1}_{x \text{ times}}) = f(1) + f(1) + \cdots + f(1) = xf(1) = xa$, and $a = f(1) = f(1 \cdot 1) = f(1)f(1) = a \cdot a = a^2$ so $k = k^2$.

For the second part: $g(f(a + b)) = g(f(a) + f(b)) = g(f(a) + f(b)) = g(f(a)) + g(f(b))$, and $g(f(ab)) = g(f(a)f(b)) = g(f(a))g(f(b))$. So yes $g \circ f$ is a homomorphism.

2.7 $2^2 \neq 2 \bmod 8$, so it is not a ring homomorphism by exercise 2.6. Alternatively, $f(3 \cdot 3) = 9 \cdot 2 = 2 \bmod 8$, but $f(3)f(3) = 6 \cdot 6 = 4 \bmod 8$ so $f(3 \cdot 3) \neq f(3)f(3)$.

2.8 Yes, it is a ring homomorphism by exercise 2.6 since $6^2 = 6 \bmod 30$. Alternatively, $f(a + b) = 6(a + b) = 6a + 6b$, and $f(ab) = 6(ab) = 6 \bmod 30 \cdot (ab) \bmod 30 = 36 \bmod 30 \cdot (ab) \bmod 30 = 6a \cdot 6b \bmod 30 = f(a)f(b)$.

2.13 Suppose that $f\colon \mathbb{Q} \to \mathbb{Z}$ is an isomorphism. Then f is homomorphic, and $f(1) = f(\frac{1}{2} \cdot 2) = f(\frac{1}{2})f(2)$. Since f is an isomorphism it must also be onto; so by theorem 2.1 (part 7) $f(1) = 1$. This implies that $f(\frac{1}{2})f(2) = 1$. f maps to $\mathbb{Z}$ so $f(\frac{1}{2}) = a$ and $f(2) = b$ for some integers a and b. But this implies that $ab = 1$ which is only true for integers $a = b = 1$. A contradiction because then $f(1) = f(\frac{1}{2}) = 1$, and f is not one-to-one.

Similarly we can show that $\mathbb{R}$ is not isomorphic to $\mathbb{Z}$.

2.14 f cannot be an isomorphism because f is not one-to-one. $f(0) = f(n) = 0 \bmod n$. Alternatively, $|\mathbb{Z}| \neq |\mathbb{Z}_n|$ so f cannot be one-to-one by the pigeon hole theorem. So there is no isomorphism between them.

2.15 Suppose that $m > n$ then $m = |\mathbb{Z}_m| > |\mathbb{Z}_n| = n$. Then any function $f\colon \mathbb{Z}_m \to \mathbb{Z}_n$ cannot be one-to-one by the Pigeon hole principle. If $n < m$ then f cannot be onto, so f can be an isomorphism only if $n = m$.